Do LLMs Interact Like Humans? Comparing Endogenous Interaction Dynamics Across Nine Social Domains

C. Ben Gibson[1]

[1] *[Department, Institution, City, Country]*

May 26, 2026

Corresponding author: C. Ben Gibson, *[email@institution.edu]*

Abstract

Large language models (LLMs) are increasingly deployed as social simulation agents, yet it remains unclear whether they reproduce the structural dynamics that govern human interaction sequences. We evaluate two LLMs—Claude Haiku 4.5 and Llama 3.1 70B, each operating as a multi-agent network participant— against empirical human interaction sequences spanning nine social domains (international conflict, classroom interaction, gang violence, email, face-to-face glances, private messaging, SMS, Twitter, and phone calls), using nine endogenous statistics drawn from the relational event model literature as benchmarks. LLMs systematically over-initiate contact (*SndInitiation*: +31%), over-repeat recent interactions (*Repetition*: +97% for Claude), over-concentrate on recently-active receivers (*RecRecencyRank*: +61%), spread interactions across too many distinct partners (*OutDegSnd*: +189% to +227%), and over-imitate the targeting patterns of other actors (*Imitation*: +43% to +48%). The one consistent deficit is in interaction-order violation (*IOviolation*), where LLMs are 25% less likely than humans to break turn-taking norms, reflecting excessive conversational deference. Llama 3.1 70B shows larger biases than Claude Haiku 4.5 on most statistics. Overall RMSE is 1.66 for Claude and 1.94 for Llama. Bias magnitudes are largest in high-volume, low-structure settings (SMS, phone) and smallest in adversarial or formal contexts (war, email). These systematic biases have implications for the use of LLMs as social simulation agents and raise questions about what aspects of social dynamics are encoded in transformer pretraining.

Keywords: large language models; social simulation; relational event models; interaction dynamics; agent-based modeling; endogenous statistics; social behavior

1 Introduction

A growing body of work uses large language models (LLMs) as computational proxies for human social behavior. LLM agents have been deployed to simulate opinion dynamics [Argyle et al., 2023], political deliberation [Manning et al., 2024], jury decision-making, and economic games [Horton, 2023, Aher et al., 2023], with researchers variously concluding that LLMs are plausible synthetic survey respondents, strategic actors, and social reasoners. A parallel literature in organizational science and computational social science employs LLMs as interacting agents in network simulations, leveraging their world knowledge to generate scenario-conditioned behavior at scale. Both applications rest on an implicit empirical claim: that LLM behavior, at least in the relevant domain, resembles human behavior closely enough to serve as a valid substitute.

This claim has received less direct empirical scrutiny than the enthusiasm surrounding LLM social simulation might suggest. Most evaluations focus on macro-level outcomes—whether aggregate opinion distributions match survey data, whether strategic equilibria are reproduced, whether argumentation patterns resemble human debates. Micro-level sequencing—the fine-grained, moment-to-moment dynamics of who interacts with whom, in what order, and how that pattern evolves over time—has received much less attention. Yet these dynamics are precisely what social network researchers study, and they are what matters most for simulation applications where the quality of simulated trajectories, not just their endpoints, is the object of interest.

Relational event models (REMs) [Butts, 2008] offer a principled framework for characterizing interaction sequences. A REM represents the probability of each next event as a function of endogenous statistics computed from the prior event history: whether the sender has previously contacted the receiver (dyadic persistence), whether the receiver has recently targeted the sender (reverse-dyad recency), whether turn-taking norms are satisfied (sequential ordering), and how broadly the sender has spread their contacts (outdegree). These statistics capture behavioral tendencies that are both theoretically interpretable and empirically robust across a wide range of social settings [Gibson, 2025]. If LLM agents are adequate models of human social actors, their interaction sequences should reproduce these statistics.

We provide a direct empirical test. We deploy two LLM providers—Claude Haiku 4.5 (Anthropic) and Llama 3.1 70B (Meta), both accessed via AWS Bedrock—as multi-agent participants in relational event sequences seeded from nine empirical corpora. Each agent is prompted with a role identity and recent event history and asked to choose its next interaction target. We compare the resulting interaction sequences to the human empirical baseline on nine endogenous REM statistics and compute per-statistic bias and overall RMSE.

Our main findings are threefold. First, LLMs systematically over-initiate, over-repeat, and over-concentrate on recently-active partners relative to humans: they behave as if endogenous pull—the gravitational force of recent interaction history—is stronger for them than it is for people. Second, the one exception is interaction-order violation: LLMs under-break turn-taking norms, indicating excessive conversational deference relative to human baselines. Third, Llama exhibits larger biases than Claude on most statistics, though both models err in the same direction for every statistic.

These findings contribute to three literatures. For social simulation researchers, they provide a calibration benchmark: specific statistics where LLM agents require correction before their output can be treated as a realistic substitute for human interaction data. For REM

researchers, they suggest that LLMs have implicitly learned something about endogenous interaction structure from their training data, but have over-weighted certain mechanisms and under-weighted others. For AI researchers, the pattern of biases raises questions about what aspects of social behavior are encoded in transformer pretraining and how prompting strategies might mitigate systematic distortions.

2 Background

2.1 Endogenous Structure in Relational Event Sequences

The relational event model [Butts, 2008] represents directed social interaction as a continuous-time point process in which the hazard that actor i directs an event to actor j at time t depends on the full prior event history:

$$\lambda_{ijt} = \lambda_0(t) \exp\Big(\boldsymbol{\beta}^\top \mathbf{s}_{ijt}\Big),$$

where $\mathbf{s}_{ijt}$ is a vector of endogenous statistics computed from events up to time t. The framework has been applied to email and phone communication [Vu et al., 2017], hospital patient transfers [Liang and Fu, 2014], organizational deliberation [Kitts et al., 2017], gang conflict [Duxbury and Haynie, 2021, Gravel et al., 2023], and international conflict [Renshaw and Gibson, 2023].

Recent work by Gibson [2025] applied greedy forward model selection to 217 empirical interaction sequences across nine social domains and identified four statistics that appear in the first four selection steps in over 60% of networks: *SndInitiation* (whether the sender has ever previously contacted the receiver), *IOviolation* (indicator of participation-shift violations), *Repetition* (sender's normalized prior send-share to the receiver), and *RecRecencyRank* (rank-based recency of the most recent reverse-dyad event). These four terms, representing dyadic persistence, sequential ordering, and reverse-dyad recency, constitute a robust baseline that generalizes across ecological contexts from private messaging to gang violence. Additional statistics capturing sender breadth (*OutDegSnd*), preferential attachment (*Imitation*), transitive and cyclic closure (*NOTPSnd*, *NITPSnd*), and weighted reciprocity (*WeightedRec*) complete the candidate set evaluated here.

An ecological distinction that shapes which endogenous statistics dominate in any given setting is the *visibility* of interaction histories: whether participants can observe each other's interaction behavior [Gibson, 2025]. In high-visibility settings (classrooms, public conflict, Twitter), imitation and reverse-dyad recency are more predictive because others' targeting behavior is observable. In low-visibility settings (private email, phone, SMS), actors can access only their own interaction history, so dyadic persistence and sequential norms dominate. This ecological distinction provides an organizing framework for interpreting cross-domain variation in LLM bias.

2.2 LLMs as Social Simulation Agents

The use of LLMs as synthetic social agents has grown rapidly since the release of instruction-tuned models capable of sustained role-playing. Early work established that LLMs can reproduce statistical patterns in survey responses when prompted to adopt specific demographic

profiles [Argyle et al., 2023], though subsequent work documented systematic biases and over-homogenization [Santurkar et al., 2023, Park et al., 2024]. In strategic interaction settings, LLMs demonstrate economically rational behavior in games like the ultimatum game and prisoner's dilemma, though with characteristic deviations from Nash equilibria [Horton, 2023]. Generative agent architectures [Park et al., 2023] have extended LLM simulation to multi-agent settings, with agents maintaining memories, forming social relationships, and coordinating behavior over extended interaction sequences.

Work specifically addressing whether LLMs reproduce the structural dynamics of social interaction sequences is sparse. Manning et al. [2024] examines conversational dynamics in LLM deliberation, and several papers in the computational social science literature use LLMs to generate synthetic network data, but systematic evaluation against the endogenous statistics that REM researchers use as behavioral benchmarks has not been conducted.

An important distinction in role-based LLM simulation is between *sender roleplay*—prompting the model to adopt a specific actor's identity and history—and *prediction*—asking the model to forecast what a hypothetical actor would do. Sender roleplay substantially outperforms prediction for reproducing empirical REM statistics in conflict sequences, suggesting that identity assignment, not just history exposure, drives accuracy in social simulation tasks [Gibson, 2024]. The multi-agent architecture evaluated here employs sender roleplay across all conditions.

3 Methods

3.1 Empirical Corpora

The evaluation draws on empirical interaction sequences from nine social domains:

- **War** (international conflict): Militarized Interstate Disputes (MID), directed conflict acts among states. High-visibility, conflictual.
- **Classroom**: Directed interactions in classroom settings. High-visibility, affiliative.
- **Gangs**: Gang violence sequences, directed aggressive acts. High-visibility, conflictual.
- **Twitter**: Directed mention and reply sequences. High-visibility, affiliative.
- **Phone**: Phone call sequences. High-visibility for some institutional settings (DoD military exercise), affiliative.
- **Email**: University email sequences. Low-visibility, affiliative.
- **ftf (face-to-face)**: Directed gaze / glance sequences. Low-visibility, affiliative.
- **PMs (private messages)**: UCI private message sequences. Low-visibility, affiliative.
- **SMS**: Short message service sequences. Low-visibility, affiliative.

A total of 27 networks across the nine domains are included in the multi-domain evaluation (between one and five networks per domain). For each network, human interaction sequences serve as the empirical baseline against which LLM-generated sequences are compared.

3.2 LLM Multi-Agent Architecture

We employ a multi-agent simulation architecture in which LLM agents operate as network participants, selecting interaction targets from the set of available actors. Each agent is prompted with a role identity (sender assignment) and a recent event history window, and asked to nominate its next interaction target. Sender activation is governed by the hazard surface of an empirically-fitted relational event model, concentrating event production on actors whose prior interaction history places them at elevated hazard. This architecture combines structural discipline from the REM with identity-conditioned receiver selection from the LLM.

Two LLM providers are evaluated in parallel:

- **Claude Haiku 4.5** (Anthropic), accessed via AWS Bedrock
- **Llama 3.1 70B** (Meta), accessed via AWS Bedrock

For each empirical network, the LLM simulation is initialized from the observed event history and run forward to produce a synthetic sequence of the same length. REM sufficient statistics are then computed over the generated sequence and compared to the empirical baseline.

3.3 Endogenous Statistics

Nine endogenous statistics from the REM candidate set of Gibson [2025] are evaluated. Each is normalized to $[0, 1]$ within sender at each event time. We describe each statistic, its behavioral interpretation, and the direction of human baseline values in Table 1.

Table 1: Endogenous statistics evaluated, organized by conceptual group.

Term	**Group**	**Description**
SndInitiation	Forward-dyad history	Binary indicator: sender i has ever previously contacted receiver j
Repetition	Forward-dyad history	i's normalized prior send-share to j; depth of prior contact
RecRecencyRank	Reverse-dyad history	Rank-recency of the most recent $j \to i$ event (1/rank)
WeightedRec	Reverse-dyad history	Squartini weighted reciprocity ratio: $\min(w_{ij}, w_{ji}) / \max(w_{ij}, w_{ji})$
Imitation	Popularity/closure	j's normalized indegree from senders other than i (preferential attachment)
OutDegSnd	Popularity/closure	i's normalized contact breadth (sender out-degree)
NOTPSnd	Popularity/closure	Normalized transitive two-path count ($i \to k \to j$)
NITPSnd	Popularity/closure	Normalized cyclic two-path count ($j \to k \to i$)
IOviolation	Sequential ordering	Indicator of a participation-shift violation (PSAB-XA, PSAB-XY, or PSAB-XB)

3.4 Evaluation Metrics

For each network and each statistic, we compute the *bias* as the difference between the LLM-generated mean and the empirical mean:

$$\text{bias}_k = \bar{s}_k^{\text{LLM}} - \bar{s}_k^{\text{emp}},$$

where $\bar{s}_k$ is the mean value of statistic k over all events in the sequence. Positive bias indicates the LLM over-produces the corresponding pattern relative to humans; negative bias indicates under-production. Mean absolute bias across statistics and networks provides an overall calibration measure.

Root mean squared error (RMSE) across statistics is computed for each network as:

$$\text{RMSE} = \sqrt{\frac{1}{K}\sum_{k=1}^{K}\left(\bar{s}_k^{\text{LLM}} - \bar{s}_k^{\text{emp}}\right)^2},$$

where $K = 9$ is the number of statistics evaluated. RMSE is reported by domain (averaged over networks within domain) and overall.

4 Results

4.1 Overall Calibration

Table 2 summarizes the grand mean and mean absolute bias for each LLM across all nine statistics, and the overall RMSE averaged across all domains and networks. Claude Haiku 4.5 (mean absolute bias: 0.74) is better calibrated than Llama 3.1 70B (mean absolute bias: 0.87) overall, a pattern that holds across most individual statistics.

Table 2: Overall calibration summary. Mean bias (positive = over-production relative to human empirical), mean absolute bias, and RMSE across all 9 statistics and 27 networks.

Model	**Mean bias**	**Mean —bias—**	**RMSE**
Claude Haiku 4.5	+0.650	0.737	1.660
Llama 3.1 70B	+0.798	0.874	1.943

The positive overall mean bias confirms that the dominant failure mode for both LLMs is over-production of endogenous structure relative to human baselines: LLMs interact with the right people too often, too repeatedly, and too responsively.

4.2 Statistic-Level Comparisons

Table 3 reports the empirical mean, LLM-generated means, and bias (absolute and percentage) for each statistic, averaged across all nine domains. The statistics are sorted by magnitude of the Claude absolute bias.

Table 3: Mean endogenous statistic values: human empirical vs. LLM-generated, averaged across all domains and networks. % bias is $(m_{\text{LLM}} - m_{\text{emp}})/m_{\text{emp}} \times 100$.

Term	Empirical	Claude		Llama	
		Mean	% bias	Mean	% bias
OutDegSnd	2.399	6.935	+189%	7.849	+227%
Repetition	0.329	0.647	+97%	0.474	+44%
RecRecencyRank	0.387	0.622	+61%	0.615	+59%
Imitation	0.563	0.808	+43%	0.833	+48%
SndInitiation	0.702	0.920	+31%	0.929	+32%
NITPSnd	0.410	0.482	+18%	0.688	+68%
NOTPSnd	0.459	0.576	+25%	0.767	+67%
WeightedRec	0.277	0.289	+5%	0.392	+42%
IOviolation	0.732	0.544	−26%	0.544	−26%

OutDegSnd: massive over-breadth. The largest divergence between LLM and human behavior is in sender outdegree breadth. Human actors engage a mean of 2.4 normalized distinct contacts; Claude reaches 6.9 (+189%) and Llama 7.8 (+227%). This is the single most consistent and severe bias across all statistics: LLMs distribute their interactions across a far wider set of partners than real actors do in any domain. The disparity is visible at every network and cannot be attributed to domain-specific factors.

Repetition and RecRecencyRank: over-concentration on recent history. LLMs re-contact the same target they most recently contacted at dramatically higher rates than humans (Repetition: Claude +97%, Llama +44%). They also preferentially target actors from whom they have most recently received at elevated rates (RecRecencyRank: +61% for both models). Together, these biases suggest that LLMs over-weight the most recent part of the event history when selecting interaction targets: they are simultaneously too sticky (returning too often to their own last-contacted partner) and too responsive (reacting too strongly to being recently targeted).

Imitation and SndInitiation: over-responsiveness and over-initiation. LLMs are more likely than humans to target actors that other senders are currently targeting (Imitation: +43% to +48%), and more likely to initiate contact with actors they have previously contacted rather than exploring new relationships (SndInitiation: +31% to +32%). The SndInitiation bias means that a larger share of LLM events are continuations with established partners, consistent with the Repetition over-production.

IOviolation: systematic under-violation of turn-taking norms. IOviolation is the only statistic where LLMs *under-produce* relative to humans. Humans violate sequential participation norms (the current addressee should respond next) in approximately 73% of events, reflecting the spontaneous, asymmetric quality of real social interaction—actors initiate new threads, pivot to new targets, and break conversational rhythm constantly. LLMs do so in only 54% of events, a 26% deficit. Both models err in the same direction, and the bias is

nearly identical (Claude: −0.110; Llama: −0.130), suggesting it reflects a general property of language model training rather than a provider-specific artifact. LLMs are more politely turn-taking than humans.

WeightedRec: the best-calibrated statistic. Weighted reciprocity—the balance between forward and reverse dyad volumes—is the most accurately reproduced statistic. Claude is nearly unbiased (+5%) and Llama shows moderate over-weighting (+42%). This is consistent with LLMs having accurate representations of reciprocal exchange norms, even if they misrepresent several other structural features.

Triadic closure (NOTPSnd, NITPSnd). Both transitive and cyclic closure are overproduced, with substantially larger biases for Llama (+67% and +68%) than for Claude (+25% and +18%). Human social networks show moderate transitivity; LLMs disproportionately favor targets with shared social connections, consistent with an over-reliance on common-neighbor heuristics in target selection.

4.3 Domain-Level Variation in RMSE

Table 4 reports RMSE by domain and model. Error is smallest in adversarial and formally structured settings (War, Email, ftf) and largest in high-volume informal settings (SMS, Phone, PMs). Both models show substantial domain variation, indicating that LLM calibration is not uniform across social contexts.

Table 4: RMSE by domain and model (lower = better). Domains sorted by Claude RMSE. Visibility and valence classifications follow Gibson [2025].

Domain	Visibility	Valence	Networks	Claude RMSE	Llama RMSE
War	High	Conflictual	1	0.134	0.836
Email	Low	Affiliative	5	0.292	0.206
ftf	Low	Affiliative	5	0.443	0.346
Classroom	High	Affiliative	1	0.600	0.444
Gangs	High	Conflictual	5	1.015	1.081
Twitter	High	Affiliative	1	1.529	1.630
PMs	Low	Affiliative	5	2.042	2.460
Phone	High	Affiliative	1	2.288	2.609
SMS	Low	Affiliative	3	7.106	8.821
Overall			**27**	**1.660**	**1.943**

The War domain yields the lowest RMSE for Claude (0.134), suggesting that LLM agents are well-calibrated for adversarial, high-visibility interaction sequences—possibly because international conflict narratives are prominently represented in pretraining data. Email sequences are well-recovered by both models (Claude: 0.292; Llama: 0.206), consistent with the preponderance of email-like text in LLM training corpora. The SMS domain is an outlier, with RMSE of 7.1 and 8.8 for Claude and Llama respectively, driven primarily by extreme OutDegSnd over-production in short, high-volume messaging sequences where LLMs continue

to distribute attention broadly even as human actors converge tightly on a small number of partners.

4.4 Bias by Statistic and Domain

Figure ?? (see `multiagent_multidomain_bias.pdf`) displays the per-term bias for both LLMs across all domains, decomposed by valence and visibility. Several patterns emerge beyond the aggregate figures.

For IOviolation, the LLM deficit varies substantially by domain. In high-visibility affiliative settings (Classroom, Twitter), LLMs are modestly below human baselines; in low-visibility high-violation domains (ftf, Phone), the gap is larger. Empirical IOviolation rates in the War domain are very low (0.054), reflecting the extreme routinization of adversarial conflict sequences; both LLMs substantially over-produce violations in this context (Claude: 0.409, Llama: 0.330), reversing the overall pattern.

For Repetition, Claude is a larger over-producer than Llama across most domains (Claude: +97% overall vs. Llama +44%), suggesting a provider-specific tendency toward dyadic inertia in target selection. In Email and ftf networks, both models are reasonably accurate, possibly because private conversational sequences in these corpora closely resemble the dialogue structure most heavily represented in LLM pretraining.

5 Discussion

5.1 The Signature of Over-Endogeneity

Taken together, the pattern of LLM biases describes what might be called *over-endogeneity*: LLMs respond more strongly to the endogenous pull of prior interaction history than humans do. They are more likely to initiate contact with already-contacted partners, more likely to return to the most recent partner, more likely to follow the most recent reverse-dyad event, more likely to target popular actors, and more likely to follow transitive closure paths. In each case the direction of the bias is toward *more structure*—more concentration, more persistence, more responsiveness to network position.

The one exception—IOviolation—is consistent with this account. Human actors violate turn-taking norms spontaneously and frequently, reflecting the autonomous, multi-threaded character of real social interaction. LLMs, trained on dialogue data that heavily represents structured conversational exchange (question-answer pairs, back-and-forth threads), have learned an over-regularized model of conversation that over-respects participation norms. The IOviolation deficit is thus the flip side of over-endogeneity: LLMs are too responsive to the implied structure of the ongoing conversation, just as they are too responsive to prior contact patterns.

The OutDegSnd bias—LLMs engage 2–3 times as many distinct partners as humans—sits slightly apart from this narrative. Unlike Repetition or IOviolation, over-breadth does not directly reflect history-responsiveness. It may reflect an instruction-following artifact: LLMs prompted to "choose a target" in a multi-actor environment may apply a distribution-seeking heuristic (vary your targets to show you're attending to the network) that has no analog in human behavior. Alternatively, it may reflect exposure to social media and discussion forum data during pretraining, where broad multi-target engagement is normative.

5.2 Ecological Modulation

Bias magnitudes are not uniform across domains. LLMs are best-calibrated in adversarial high-visibility settings (War, Gangs) and worst-calibrated in informal high-volume low-structure settings (SMS, PMs). Several factors may account for this pattern.

First, adversarial sequences have highly concentrated interaction histories with clear retaliatory logic; the strong endogenous pull of retaliatory dynamics may be one of the features LLMs have most accurately internalized from training data, leaving less room for overproduction. Second, SMS and private messaging sequences involve very rapid, high-volume exchange between a small number of partners, and human behavior in these settings is highly constrained by the conversational dyad. LLMs, which continue to spread attention broadly regardless of context, are poorly suited to reproducing this structure.

The visibility distinction provides a partial but imperfect account. Low-visibility domains (Email, ftf) show lower RMSE than high-visibility domains on average for Llama, but Claude performs better in some high-visibility settings. The relationship between visibility and calibration likely reflects domain-specific pretraining exposure as much as an ecological mechanism.

5.3 Implications for Social Simulation

Our results suggest that LLM multi-agent simulations of human interaction sequences are subject to predictable, measurable biases that researchers should account for when using LLM-generated sequences as proxies for human behavior. The most critical corrections are:

1. **OutDegSnd**: LLMs will over-estimate how broadly actors distribute their interactions. Network density and contact breadth metrics derived from LLM simulations will be inflated by factors of 2–3.

2. **Repetition and RecRecencyRank**: LLMs will over-estimate dyadic persistence and reverse-dyad responsiveness, biasing estimates of relationship strength and reciprocity.

3. **IOviolation**: LLMs will under-estimate the spontaneity and asymmetry of real social interaction, producing sequences that are too orderly and turn-taking.

These biases are systematic and consistent across two distinct LLM architectures, suggesting they are not easily resolved by provider choice. Prompt engineering strategies that explicitly penalize over-repetition, over-breadth, and excessive turn-taking compliance—or statistical post-processing that rescales generated sequences toward empirical baselines—may be required for applications where quantitative fidelity is important.

The finding that Claude Haiku 4.5 consistently outperforms Llama 3.1 70B across most statistics (lower absolute bias, lower RMSE on 7 of 9 domains) suggests that model scale and instruction-following calibration matter for this task, though the dominant biases are shared across providers.

5.4 What LLMs Have Learned About Social Interaction

The pattern of biases is informative about the implicit social model encoded in transformer pretraining. LLMs have apparently learned that social interaction tends to be:

- Repeated: established contacts are more likely next contacts.
- Responsive: being targeted elevates the probability of targeting back.
- Orderly: conversations proceed in structured turn-taking sequences.
- Broad: actors in social networks maintain relationships with many partners simultaneously.
- Popular-target-seeking: high-indegree actors attract additional interaction.

The first two of these are directionally correct but over-calibrated. The third is directionally incorrect: LLMs over-respect turn-taking norms, while humans frequently violate them. The fourth reflects genuine human behavior in some settings (broadcast media, social media) but not in the dyadic, private settings where humans are most tightly concentrated. The fifth is weakly correct: preferential attachment is a real tendency in human networks, but LLMs over-weight it.

These characterizations suggest that LLMs have learned a stylized, broadcast-media model of social interaction that fits public discourse data (Twitter, news commentary, structured debate) better than private dyadic interaction data. The mismatch is largest in SMS and private messaging—the settings furthest from public discourse—and smallest in war and email, where the relevant interaction structure is closer to the formal, asymmetric exchanges well-represented in pretraining.

5.5 Limitations

Several limitations bear on interpretation. First, the multi-agent simulation architecture governs sender activation via the empirical REM hazard surface, which may impose structural constraints that partially counteract LLM-native biases. Purely unconstrained LLM simulation might show different (possibly larger) biases. The biases we document are thus lower bounds on the calibration gap under fully endogenous conditions.

Second, the LLMs are evaluated in English with generic role prompts. Domain-specific prompting—providing LLMs with detailed sociological context about gang interaction norms, classroom dynamics, or SMS conventions—might substantially improve calibration. Whether the biases we document reflect fundamental properties of LLM social models or prompt-sensitivity artifacts is not determined by the present study.

Third, our evaluation uses nine statistics from one candidate set; additional endogenous mechanisms not in this set may be more or less well-reproduced. The 97% AIC recovery reported for this term set in Gibson [2025] suggests it captures the dominant endogenous structure, but LLM biases on unrepresented statistics cannot be ruled out.

Fourth, both models are accessed through the same API infrastructure (AWS Bedrock), and any systematic effects of the API wrapper, quantization, or inference configuration would affect both providers equally. Direct comparison of provider-specific biases should be interpreted with this in mind.

6 Conclusion

We have shown that LLM multi-agent participants systematically over-produce most endogenous interaction statistics relative to human empirical baselines while under-producing spontaneous norm violation. Across nine social domains, two LLM providers (Claude Haiku 4.5 and Llama 3.1 70B) consistently show the same directional biases: too much contact breadth, too much dyadic repetition, too much reverse-dyad responsiveness, too much imitation of popular targets, and too little turn-taking violation. The magnitudes are largest in informal high-volume settings (SMS, phone) and smallest in adversarial or formally structured settings (war, email). Claude is better calibrated than Llama overall, but both share the same failure modes.

These findings suggest that LLMs have learned a systematically distorted model of social interaction that over-weights endogenous structural tendencies and under-weights the spontaneous, asymmetric quality of real human behavior. For researchers using LLM agents as social simulation proxies, these biases are measurable, consistent, and in principle correctable—but they must be acknowledged and accounted for before LLM-generated sequences are treated as valid substitutes for human interaction data.

Acknowledgements

[Insert acknowledgements here, or state: The author received no specific funding for this work.]

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

The empirical sequences used in this study are drawn from multiple previously published corpora. *[Specify dataset availability.]* Analysis code and generated sequences are available at `https://github.com/\ldots`.

References

Gati V. Aher, Rosa I. Arriaga, and Adam Tauman Kalai. Using large language models to simulate multiple humans and replicate human subject studies. *Proceedings of the 40th International Conference on Machine Learning*, pages 337–371, 2023.

Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351, 2023.

Carter T. Butts. A relational event framework for social action. *Sociological Methodology*, 38 (1):155–200, 2008.

Scott W. Duxbury and Dana L. Haynie. Criminal network security: An agent-based approach to evaluating network resilience. *Criminology*, 59(1):5–33, 2021.

C. Ben Gibson. Do LLMs know how wars work? role assignment and endogenous structure in LLM-simulated conflict sequences. Working paper, 2024.

C. Ben Gibson. These four first: A robust endogenous baseline for relational event sequences. Working paper, 2025.

Jason Gravel, George E. Tita, and Avinash S. Bhati. Rivalries, retaliation, and the temporal dynamics of gang violence. *Journal of Quantitative Criminology*, 39:155–184, 2023.

John J. Horton. Large language models as simulated economic agents: What can we learn from homo silicus? *NBER Working Paper*, (31122), 2023.

James A. Kitts, Alessandro Lomi, Daniele Mascia, Francesca Pallotti, and Eric Quintane. Investigating the temporal dynamics of interorganizational exchange: Patient transfers among italian hospitals. *American Journal of Sociology*, 123(3):850–910, 2017.

Hai Liang and King-wa Fu. Testing propositions derived from twitter studies: Generalization and replication in computational social science. *PLOS ONE*, 9(8):e115983, 2014.

Benjamin S. Manning, Kehang Zhu, and John J. Horton. Automated social science: Language models as scientist and subjects. *NBER Working Paper*, (32381), 2024.

Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pages 1–22, 2023.

Peter S. Park, Jakob Goldsmith, Rylan Schaeffer, and Sanmi Koyejo Sun. The generative social science of large language models. *arXiv preprint*, 2024. arXiv:2406.07712.

Scott Renshaw and C. Ben Gibson. Modeling conflict dynamics with relational event models. *Journal of Peace Research*, 60(4):700–714, 2023.

Shibani Santurkar, Esin Durmus, Faisal Ladd, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. Whose opinions do language models reflect? *Proceedings of the 40th International Conference on Machine Learning*, pages 29971–30004, 2023.

Duy Q. Vu, Alessandro Lomi, Daniele Mascia, and Francesca Pallotti. Relational event models for longitudinal network data with an application to interhospital patient transfers. *Statistics in Medicine*, 36(14):2265–2287, 2017.